

Hacking law firms with abandoned domain names

Hacking law firms with abandoned domain names - Gabor, @gszathmari, Rainbow and Unicorn.

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Content

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Email is an essential service for all businesses, including legal practices. Email is not only a primary communication channel but also required for registering with online services and profession-specific portals. When law firms merge or wind-up, internet domain names are often abandoned, allowing anyone to re-register and take ownership of the former firm's domain name. The new owner can then, among other things take control of the former firm's email services.

This research report demonstrates how domain name abandonment attacks pose a significant cyber threat to the legal profession and other businesses. This report also makes recommendations as to measures legal practices and other businesses can take to stop this threat.

Gabor Szathmari is a cybersecurity expert with over ten years experience, having worked in both private and public sectors. He has helped numerous big-name clients with data breach investigations and security incident management. In his professional life, Gabor helps businesses, including many small and mid-size legal practices, with their cybersecurity challenges at [Iron Bastion – Australia's anti-phishing experts](#).

Update (18/09/2018): Read the high-level summary of this research on [Iron Bastion's security blog](#).

Update (12/09/2018): Our slides from SecTalks Sydney are available [here](#).

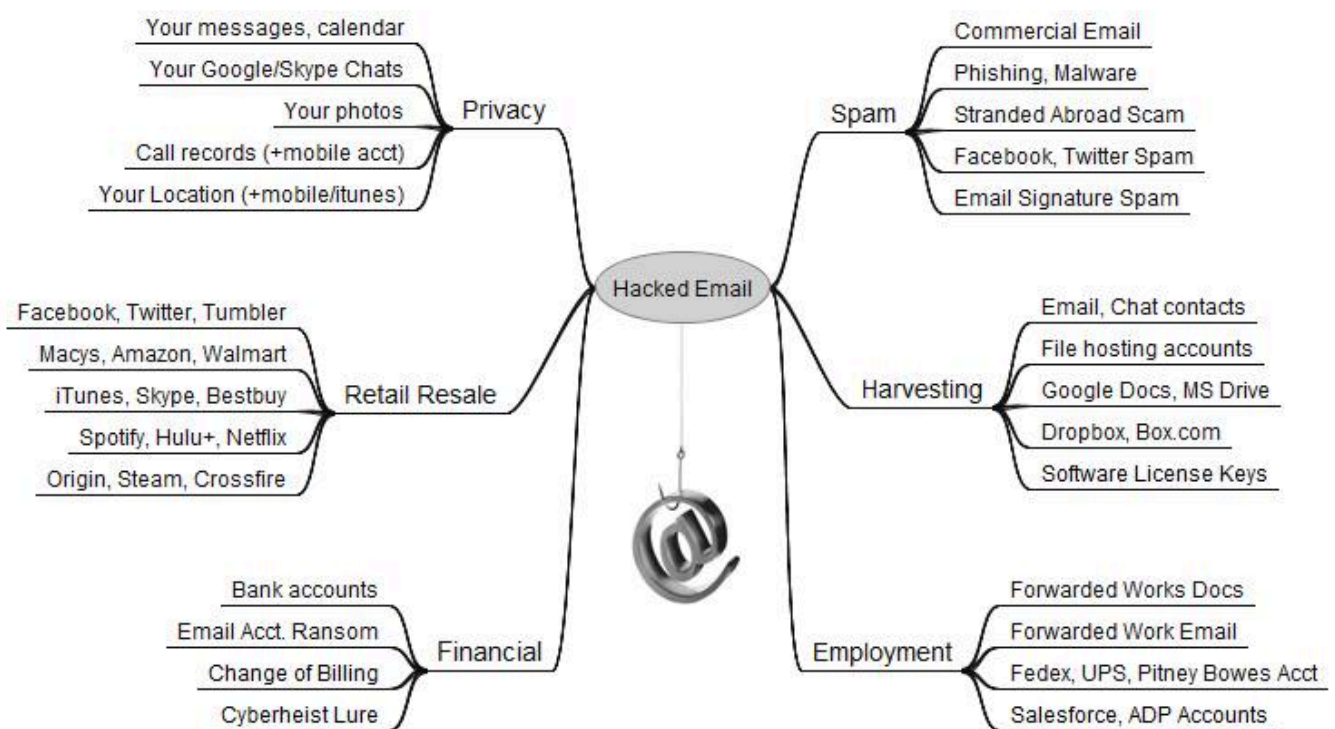
Domain name abandonment allows cybercriminals to gain access to, or reset passwords for online services and profession-specific portals. These online services store documents, emails and other information relating to a legal practice, including financial details, personal information, confidential information and client-legal privileged information.

The goal of this research is to raise awareness of a common practice in the legal profession, and in other business of allowing domain names to expire after mergers and acquisitions. We give practical tips at the conclusion of this report on how legal practices and technology providers can defend legal practices and other businesses from domain name abandonment attacks.

A domain name is the foundation of every business

Email is an essential service in every business, and the effect of a company losing control over their email service is devastating, even if the company has merged or shut down.

Sensitive information and documents are often exchanged over emails between clients, colleagues, vendors and service providers due to the convenience. Consequently, if a bad actor takes control of an entire business's email service, sensitive information can end up in wrong hands.



The value of a hacked email account (Source: krebsonsecurity.com)

Email besides being used for communication is commonly required for **signing up for online services**. People often change jobs and end up with multiple user accounts on these services, with the old user accounts often abandoned. Online services usually rely on a single factor to reset passwords, i.e. only an email address is required to regain access if the password is forgotten. Consequently, whoever has control over the domain and able to set up a basic email service can capture password reset emails.

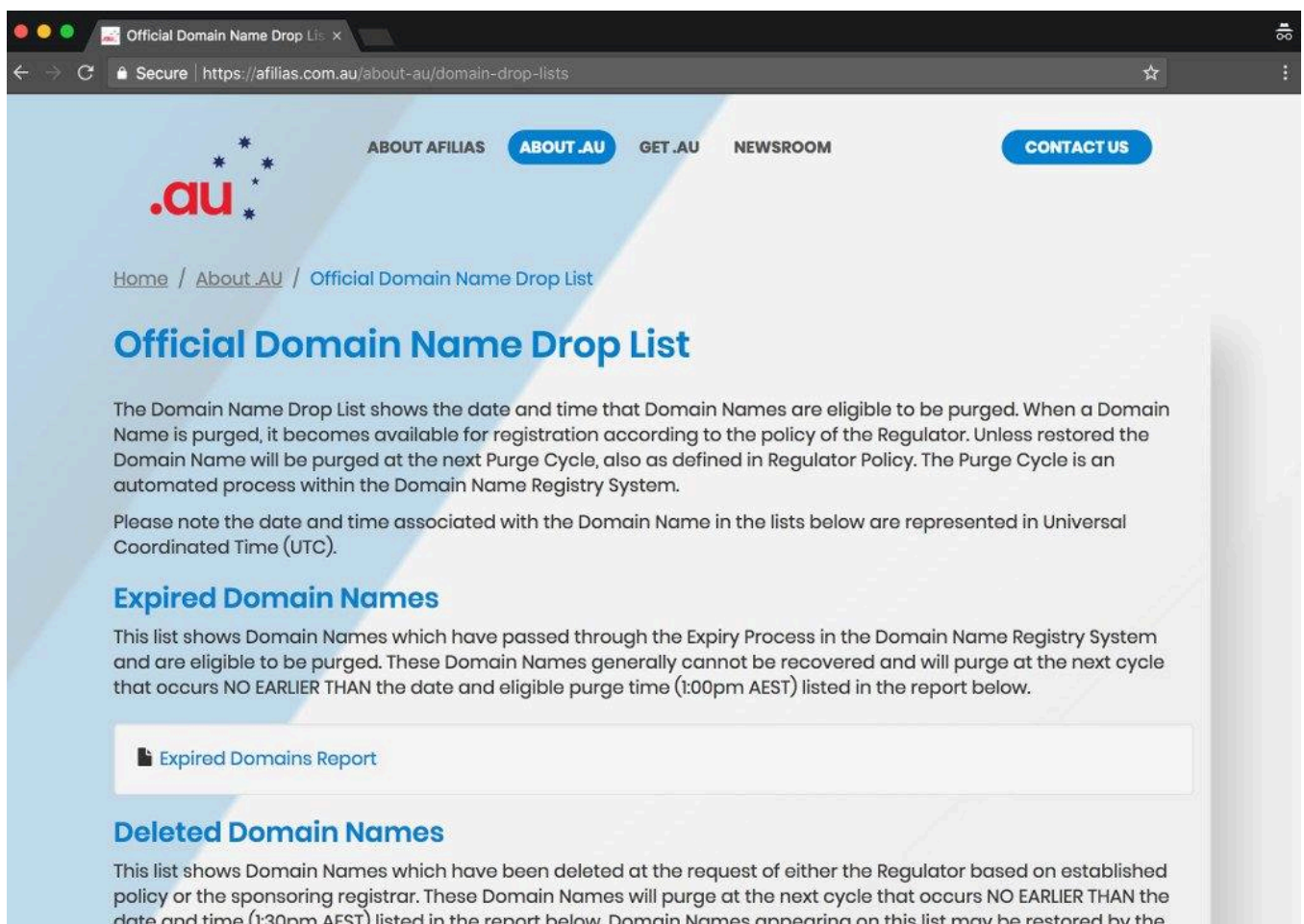
In short, bad actors can re-register an abandoned domain of a business and take full control of email services configuring it to:

- **receive email correspondence sensitive in nature; and**
- **use the email accounts to reset passwords to online services.**

What happens when a domain name expires

Once someone stops paying for an internet domain name, the registration status of the domain goes through [various stages before it gets deleted](#). Once the final grace period ends, the internet domain name is abandoned. In other words, the domain name of the former business becomes available for anyone to re-register, with no additional identity or ownership verification required. Domain registration of abandoned domains is a well-known technique amongst [SEO professionals](#) and [spam trap operators](#), but not so well-known to cybersecurity professionals as a security risk.

On any given day, an average of about a thousand '.au' domain names expire. The '.au' being the country code Top Level Domain (ccTLD) for Australia. The [list of expiring internet domain names is public](#) and published on a daily basis in a simple CSV file format. This list allows you to watch for valuable domain names due to expire and register them once the domain name registrar drops them.

A screenshot of a web browser displaying the 'Official Domain Name Drop List' page. The browser's address bar shows the URL 'https://afilias.com.au/about-au/domain-drop-lists'. The website has a blue header with the '.au' logo and navigation links: 'ABOUT AFILIAS', 'ABOUT .AU', 'GET .AU', 'NEWSROOM', and 'CONTACT US'. The main content area has a blue background with white text. It includes a breadcrumb trail: 'Home / About .AU / Official Domain Name Drop List'. The title 'Official Domain Name Drop List' is in large blue font. Below it, a paragraph explains that the list shows when domain names are eligible for purging. Another paragraph notes that dates and times are in UTC. There are two sections: 'Expired Domain Names' and 'Deleted Domain Names', both with blue headings. Each section has a brief description of the list's purpose. A button labeled 'Expired Domains Report' is visible under the 'Expired Domain Names' section.

The list of expiring domains is public

All you need to do is monitor the public list for domain names featuring relevant keywords you are interested in such as 'law' or 'legal', and register them again with your preferred domain registrar.

Once the domain registration is complete, you can specify (by changing the [MX records](#) of the domain) how the incoming emails should be handled. Having ownership of the domain name means you have full control over the incoming email flow of the former business.

By setting up a simple [catch-all email service](#), you can:

- receive email correspondence addressed to former staff; and
- receive password reset emails from online services.

Having working access to an email address is powerful because a password reset allows you to regain access to a myriad of services originally belonging to the former business and its staff.

For example:

- email platforms – Office 365, G Suite;
- shadow IT accounts signed up by individual employees for business use – particularly for file sharing – Dropbox, OneDrive, Google Drive;
- practice management software – LEAP, SILQ, ActionStep;
- legal portal software – LawConnect, GlobalX, Infotrack, VOI providers;
- online court portals – NSW Online Registry, Commonwealth Courts Portal;
- government portals – Australian Taxation Office (ATO) Business Portal;
- social media accounts – LinkedIn, Twitter, Facebook; and
- online shopping services – eBay, PayPal, Amazon.

Legal practices merge and wind-up on a regular basis

Legal practices are established and wound-up just like any other business entity on a regular basis. What makes legal practices unique is that they frequently merge with each other or are acquired by another entity and this often coincides with a name or brand change.

In the US, [2017 was a record year for top-tier law firm mergers](#) with 102 mergers or acquisitions in the year. At the small legal practice level, the number is likely to be in the thousands.

Secure | https://www.australasianlawyer.com.au/news/top-independent-firms-merge-to-challenge-australias-status-quo-245717....

Home > News

Australasian
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News Forum TV Deals Appointments Resources Events Subscriptions

Top independent firms merge to challenge Australia's status quo

AddThis by Sol Dolor | 19 Jan 2018

HWL Ebsworth and TressCox Lawyers are to combine next month, boosting what is already Australia's largest legal partnership.

The majority of the partners and staff of TressCox will join **HWL Ebsworth** on 5 February. The merger will result in the retirement of the TressCox brand, the firms have confirmed.

Both **HWL Ebsworth** and TressCox

LATEST JOBS

- Corporate/Commercial Associate - 6+ years PQE
- Senior Commercial Property Lawyer
- Corporate Solicitor - 2-4 year pqe

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The project is one of

AUSTRALASIAN LAWYER TV



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Mergers and acquisitions are also frequent in Australia

What happens after a merger or acquisition is that one entity may drop its branding in favour of the other firm, or a new brand is created for the firm. Consequently, the internet domain names of the old businesses are often left to expire in the process.

On a broader scale, [two out of three small businesses cease operating within the first three years of starting](#) according to the Australian Bureau of Statistics (ABS). This means that the domain name of many of these failed businesses is abandoned as well.

How we managed to get access to former law firms

Legal professionals also rely on emails to communicate with clients, while the staff uses their business email address to register to profession-specific legal services such as online court registries (e.g. Commonwealth Courts Portal) and other online services like Dropbox.

As part of this research, we identified a handful of abandoned domain names formerly belonging to legal practices and re-registered those domains with the intention of reinstating the email service. We set up a catch-all email server and waited for the incoming emails.

By taking full control over previously abandoned domain names, we can demonstrate that we were able to:

- access confidential documents of the former clients;
- access confidential documents of the former practice;

- access confidential email correspondence; and
- access personal information of former clients.

Also, we could have:

- impersonated legal practitioners to defraud former clients and fellow practitioners;
- regained access to the former legal practices Office 365 and G Suite account, potentially gaining access to any email and documents not deleted on the platforms; and
- hijacked personal user accounts (LinkedIn, Facebook, etc.) of the legal professionals practising in their new jobs.

Opening Pandora's Box

For this research, we hand-picked and re-registered domain names formerly belonging to legal practices in Australia. Once these domains were registered, we set up our private email server to receive emails addressed to the former legal practices.

Once the email server was ready to go, we:

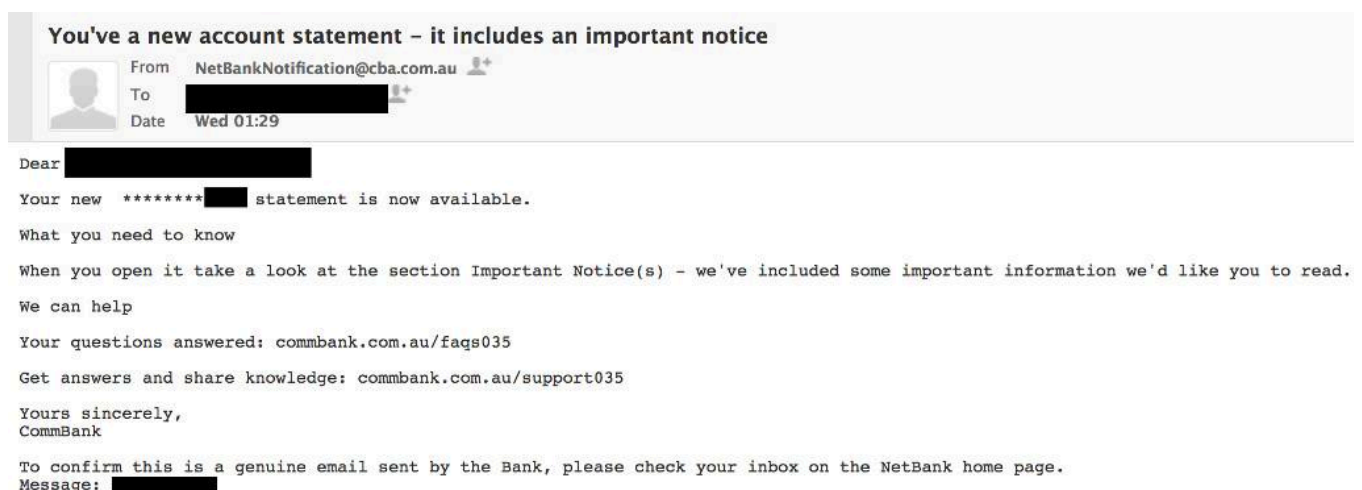
- sat back and waited for the emails to come in;
- registered the domain name to data breach websites to collect email addresses and passwords belonging to former staff; and
- attempted to reset passwords on third-party online services.

In the following sections, we are detailing what we managed to get access to and how we did it.

Emails with Sensitive Details

From the incoming emails we received, we noticed many online services send their users newsletters, reports, statements and notifications with confidential information.

We have found that NAB, Commonwealth Bank and Bankwest are popular banking services amongst legal practitioners in Australia:



From National Australia Bank <donotreply-internetbanking@nab.com.au>
Subject Hi - Your NAB Smart Statements are now available for multiple accounts

Reply Reply All Forward Archive Junk Delete More 2/7/18, 00:50



The banner features the NAB logo (a red star) in the top left corner. The main text "NAB Smart Statement" is in large, bold, white font. Below it, in smaller white text, is "View, save or print your NAB Smart Statements". To the right of the text is a stylized illustration of a computer monitor displaying a document with red and blue horizontal bars.

Dear Customer

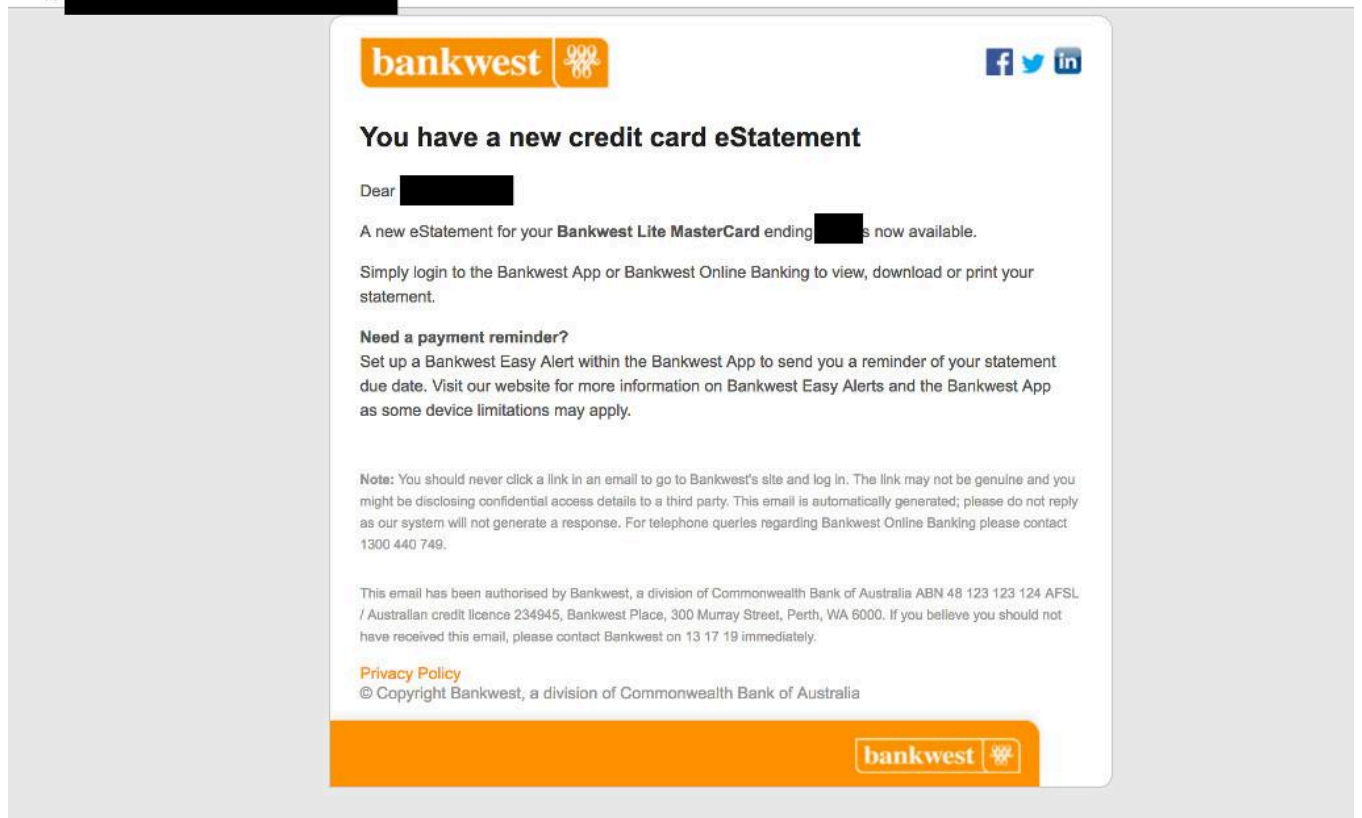
The latest NAB Smart Statement is now available for your accounts ending in:
- 6880
- 9910

To access your NAB Smart Statement in NAB Internet Banking:

1. Select 'Statements' within the 'Account' menu
2. Select your account
3. Choose 'most recent statement' OR 'search by date' then click search

From Bankwest <bob.mailbox@bankwest.com.au>
Subject Bankwest - You have a new credit card eStatement
To [REDACTED]

Reply Reply All Forward Archive Junk Delete 18/1



The email content is enclosed in a white box with an orange border. At the top left is the Bankwest logo (orange square with white text). To its right are social media icons for Facebook, Twitter, and LinkedIn. The main heading is "You have a new credit card eStatement". Below this is a salutation "Dear [REDACTED]". The body text states: "A new eStatement for your Bankwest Lite MasterCard ending [REDACTED] is now available. Simply login to the Bankwest App or Bankwest Online Banking to view, download or print your statement." It then asks "Need a payment reminder?" and provides instructions to set up a Bankwest Easy Alert. A note at the bottom states: "Note: You should never click a link in an email to go to Bankwest's site and log in. The link may not be genuine and you might be disclosing confidential access details to a third party. This email is automatically generated; please do not reply as our system will not generate a response. For telephone queries regarding Bankwest Online Banking please contact 1300 440 749." At the very bottom, it says "This email has been authorised by Bankwest, a division of Commonwealth Bank of Australia ABN 48 123 123 124 AFSL / Australian credit licence 234945, Bankwest Place, 300 Murray Street, Perth, WA 6000. If you believe you should not have received this email, please contact Bankwest on 13 17 19 immediately." and "Privacy Policy © Copyright Bankwest, a division of Commonwealth Bank of Australia". The Bankwest logo is repeated in an orange box at the bottom right.

Business debit cards often remain active even after the business has dissolved:

From NetBankNotification@cba.com.au ☆
Subject **A transaction was declined**
Reply to DoNotReply@cba.com.au ☆
To [REDACTED]

19/7/18, 12:44

Reply Reply All Forward Archive Junk Delete More

Dear customer

A \$899.64 transaction was declined on your credit card **** * as it would've gone over your chosen spending cap.

What you can do:

- Turn your spending cap off or set it to a higher amount in NetBank or CommBank app
- Keep your spending cap set to control your spending

If this wasn't your transaction, check with the additional cardholder (if you have one). If you think your card's been compromised, cancel it in NetBank by going to Settings or in the CommBank app by going to Cards> credit card> Lost stolen damaged card.

Yours sincerely,

CommBank

Travel arrangements are made on behalf of the former law firm:

From [REDACTED]
Subject [REDACTED]
To [REDACTED]
Cc [REDACTED]

13/7/18, 13:52

Reply Reply All Forward Archive Junk Delete More

Hi [REDACTED]

Please find attached your invoice from the travel agent.

A few things to note:

1. The discounted rate is subject to payment being made upfront by 25 July 2018. Normally \$450 pn, our rate \$230 pn.
2. The international transfer details are on the invoice. Conversion \$1 AUD = \$17,000 VND.
3. If payment is not received by 25 July, they will release your booking and the discounted rate will no longer be available.
4. If your internet banking does not have international transfer options, you can transfer to [REDACTED] and he can on forward (but let me know).

Thanks kindly!

Regards,

[REDACTED] Personal Assistant

Insolvency advisors

1 attachment: [REDACTED].pdf 122 KB

Save



BOOKING CONFIRMATION

YELLOW RAINBOW Co., Ltd
92 Nguyen Thai Hoc Street, Tan Phu District
Hochiminh City
www.roomrainbow.com

DATE:
REF.#
CURRENCY: USD

Contact person: Linh Trang
Title: Reservation Manager
Phone: +84 911370950
Email: info@roomrainbow.com

To:

Service **THE REVERIE SAIGON**
at: *Address:* 22-36 Nguyen Hue
Boulevard | District 1 | Ho Chi
Minh City

STAY PERIOD	ROOM TYPE	QUATITY	NUMBER OF NIGHT	RATE PER NIGHT	TOTAL
	Deluxe Double room	01	04	US\$180	US\$720
5% service charge and 10% VAT					US\$831.6
TOTAL					US\$832
TOTAL in VND (exchange rate 1USD = 23,041)					19,170,000

Legal professionals usually add their work email addresses to their current LinkedIn profile. Although because people tend to forget removing these abandoned email addresses from their profile, we keep receiving email notifications from LinkedIn:

From <messages-noreply@linkedin.com> ☆

Subject sent you an invitation on LinkedIn

To

↩ Reply

↩ Reply All

➡ Forward

📁 Archive

🗑 Junk

🗑 Delete

More

15:30

LinkedIn

wants to add you to their network

Managing Director at
Waikato, New Zealand · connections

Accept 's invitation

LinkedIn is a social network and online platform for professionals. [Learn More](#)

Former firms keep getting BAS notifications either for former clients or their former businesses:

From BASnotification@ato.gov.au ☆

↩ Reply

↩ Reply /

Subject Your next Activity Statement is now available

To [REDACTED]

ï»¿



Australian Government

Australian Taxation Office

Online Activity Statement notification

Your next online activity statements are now available for collection. A list of your registered lodgments, including the ABN and business name, is provided after the 'How to pay' section of this message.

Note: You have received this notification because your email address has been nominated to receive the 'Online activity statement'.

Online activity statements available for collection:

[REDACTED] SUPERANNUATION FUND,
[REDACTED] SUPERANNUATION FUND.

This information is current as at February 2017

The Australian Taxation Office will never send an email requesting you to confirm, update or disclose your personal details. If you receive an email like that, please do not respond to it.

© Commonwealth of Australia

Invoices sent to the legal practice can reveal which suppliers they use, the following invoice is for a legal archive storage service.

FILEMAN
COST RECOVERY SERVICE
(creditor)watch

\$45.10 due 15 Jul



FILEMAN
COST RECOVERY SERVICE
(creditor)watch[®]
member

From Fileman Pty Ltd
Pacifica Building
Suite 205
14 Bruce Avenue
PARADISE POINT QLD 4216
AUSTRALIA
T 1800 820 061
finance@fileman.com.au

We received legal documents relating to family law matters:

From: [REDACTED]
Subject: [REDACTED]
To: [REDACTED]

[Reply](#) [Reply All](#) [Forward](#) [Archive](#) [Junk](#) [Delete](#) [More](#)

9/6/18, 13:54

From: [REDACTED]
Sent: Tuesday, 1 May 2018 4:42 PM
To: [REDACTED]
Subject: [REDACTED]

Dear [REDACTED]

Please see attached letter from [REDACTED] to the Court for your records.

Kind regards

[REDACTED] Accredited Specialist (Family Law) | [REDACTED]

[REDACTED]

[REDACTED]
SOLICITORS & ATTORNEYS

[REDACTED] Sydney NSW 2000

[REDACTED]
Tel: [REDACTED]

Fax: [REDACTED]

Email: [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

CC: [REDACTED]
Sydney NSW

By email to: [REDACTED]

Dear List Clerk,

RE: [REDACTED]

Our client's application for Orders to set aside Orders made by consent by the Local Court at [REDACTED] under the provisions of s79A of the *Family Law Act 1975* has been listed for Hearing on [REDACTED]

Pursuant to what we under are the provisions of *Family Law Rule 15.12* we have filed and served our client's updated affidavit sworn [REDACTED] 2018 but we have not filed the exhibits to this affidavit. Would you please advise whether such exhibits are to be tendered at the commencement of the Hearing on [REDACTED] 2018 or whether they should be filed at an earlier time.

Yours faithfully,
[REDACTED]

Also, invoices from other law firms for work performed on behalf of the firm:



Our ref [REDACTED]
Invoice: [REDACTED]
Responsible Principal: [REDACTED]

10 May 2018

BY EMAIL: [REDACTED]

TAX INVOICE - [REDACTED]

Professional Fees

Date	Description	Fee Earner	Units	Amount (ex. GST)
			2	\$120.00
			1	\$60.00
			1	\$60.00
			1	\$60.00
			43	\$2,580.00
			1	\$60.00
Total:			49	\$2,940.00

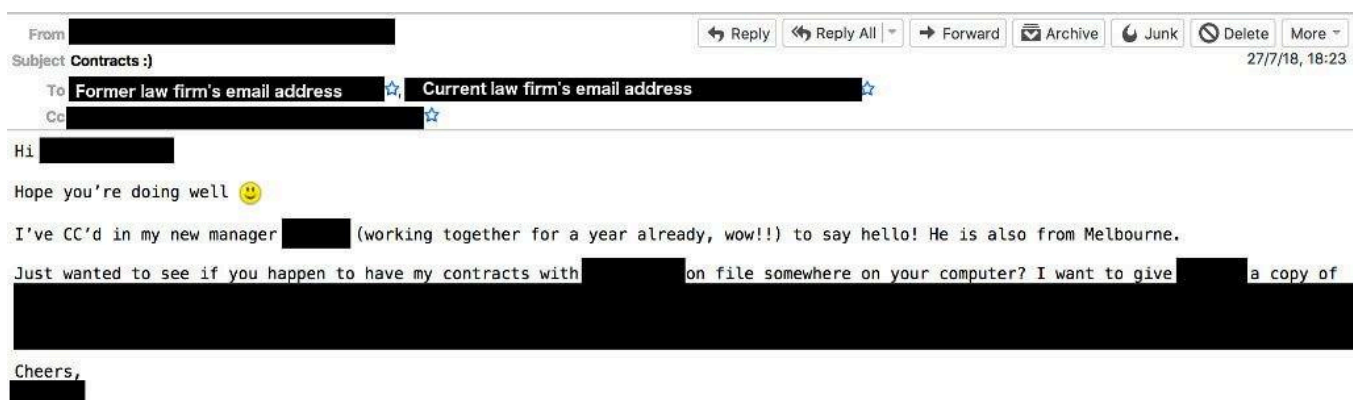
INVOICE SUMMARY

Professional Fees	\$2,940.00
GST	\$294.00
This Invoice Total (Due in 7 days)	\$3,234.00
Outstanding Invoices (Due now)	\$10,000.00
Amount we intend to pay from Trust	\$0.00
TOTAL AMOUNT DUE	\$13,234.00

We received transcripts of court proceedings:



These incoming emails let us peek into the internal workings of a law practice:



We received emails from former clients seeking advice:

From [REDACTED]
Subject [REDACTED] Divorce case
To [REDACTED]
Cc [REDACTED]

Hi [REDACTED]

Hope you are keeping well.

I don't know if you remember my case of some years ago, but you very kindly fought for me to return to UK with my son [REDACTED]

You'll be pleased to know that I have settled back in UK with [REDACTED]

[REDACTED]

I look forward to your earliest response.

Many Thanks and Kindest Regards.

[REDACTED]

[REDACTED]

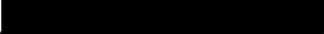
United Kingdom
[REDACTED]

Legal practitioners on the opposing sides of matters often voluntarily exposed information to us sending correspondence to the former law firm's email address as an additional cc:

From Other law firm
Subject
To Former law firm
Cc

 Reply  F

Please see attached.

 | Director
 - Family Law


OUR REF: [REDACTED]
YOUR REF:

1 August 2018

[REDACTED]
SYDNEY NSW 2000

Dear Colleagues,
[REDACTED]

Referring to the correspondences from [REDACTED] concerning [REDACTED]'s schooling for the future, would you please inform me of your client's position/proposal regarding [REDACTED]'s schooling.

Yours faithfully,
[REDACTED]

This other case involves a joint bank account closure:

From: [REDACTED]
Subject: [REDACTED]
To: [REDACTED]

 Reply  Reply All  

From: [REDACTED]
Sent: Monday, 30 July 2018 5:35 PM
To: [REDACTED]
Cc: [REDACTED]
Subject: [REDACTED]

Dear Mr [REDACTED]

[REDACTED]

We act on behalf of Mr [REDACTED] and hereby *enclose* our correspondence for your attention.

We have also carbon copied Ms [REDACTED]'s Solicitor to this email.

If you have any queries, please do not hesitate to contact our office.

[REDACTED]
Senior Paralegal

30 July 2018

Dear Mr [REDACTED]
[REDACTED]

We refer to the above and to the Joint account ending [REDACTED] with the National Australia Bank in the name of our Client and Ms [REDACTED]. We hereby **enclose** a copy of a bank statement for the account for your attention.

Accordingly we are instructed to seek that the following occur forthwith:

1. That the NAB Account ending [REDACTED] be closed; and
2. That any remaining funds be held in [REDACTED] Trust Account on behalf of the Parties.

We look forward to your confirmation that the above has occurred.

If you have any queries in relation to the above, please do not hesitate to contact our office.

Yours sincerely,



NAB Classic Banking

For further information call 13 22 65 for Personal Accounts or 13 10 12 for Business Accounts.

Account Balance Summary

Opening balance
Total credits
Total debits
Closing balance

Statement starts 18 January 2018
Statement ends 17 July 2018

Outlet Details

Account Details

Lending Investment & Insurance Enquiries

Banker
Telephone number

BSB number
Account number

Transaction Details

Date	Particulars	Debits	Credits	Balance
	Brought Forward			
	Settlement Surplus			

This document details the negotiation strategy of a settlement:

From [REDACTED]

Subject: FW: Financial Separation Negotiations

To [REDACTED]

Dear [REDACTED]

↩ Reply

↩ Reply All

➡ Forward

📁 Archive

🗑 Junk

🗑 Delete

More ▾

18:47

I hope this email finds you well ...

You will see in your diary that I have booked you for a 1 hour teleconference tomorrow - that should give us enough time for you to get up to speed on what is happening and to help formulate a strategy going forward. [REDACTED]

[REDACTED]

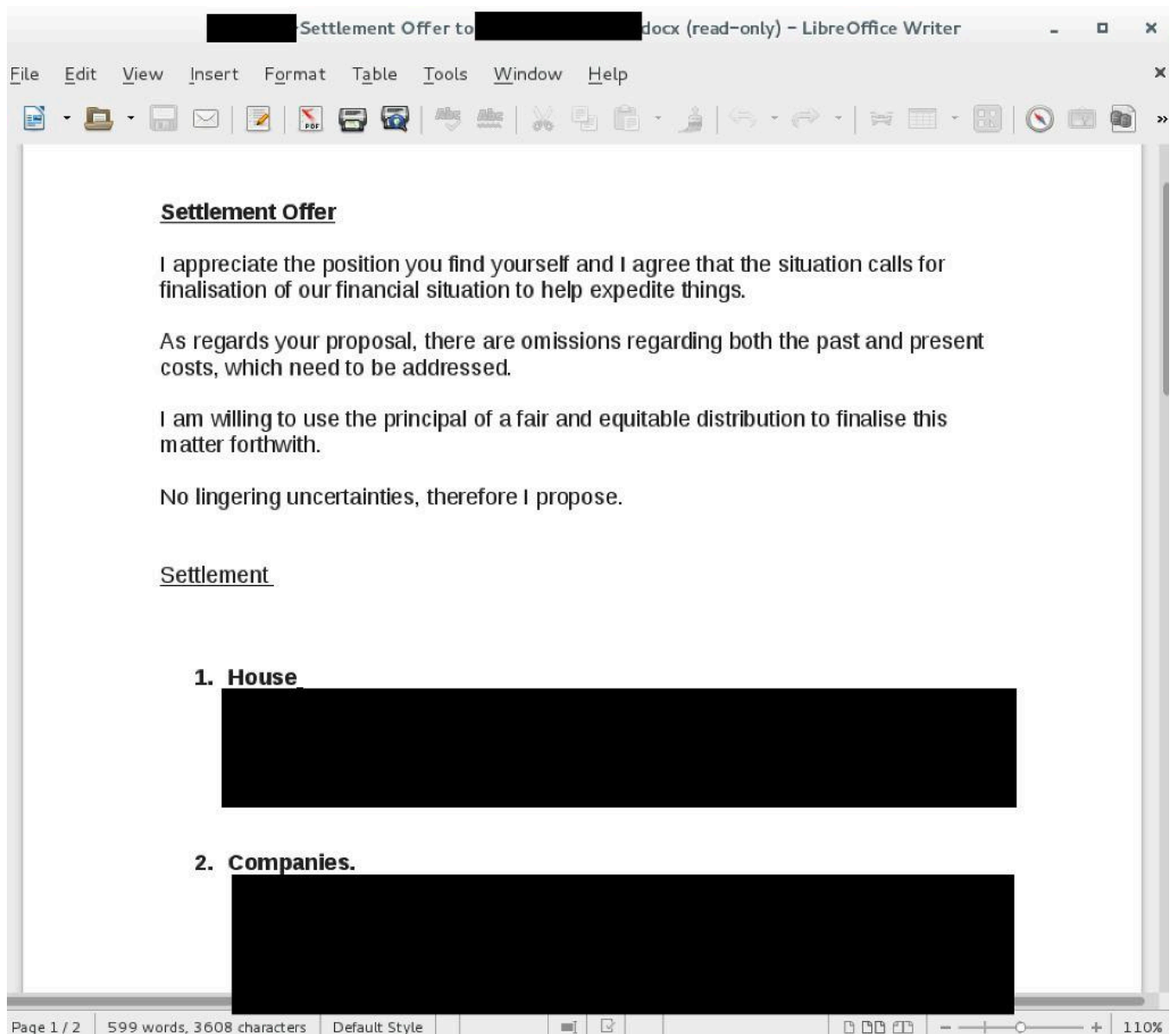
I will call your office to discuss. Speak soon.

Regards,

[REDACTED]

▶ 1 attachment: [REDACTED] Settlement Offer to [REDACTED].docx 17.9 KB

↓ Save ▾



Other Amusing and Fun Facts

Uber is the preferred choice of travel amongst legal practitioners:

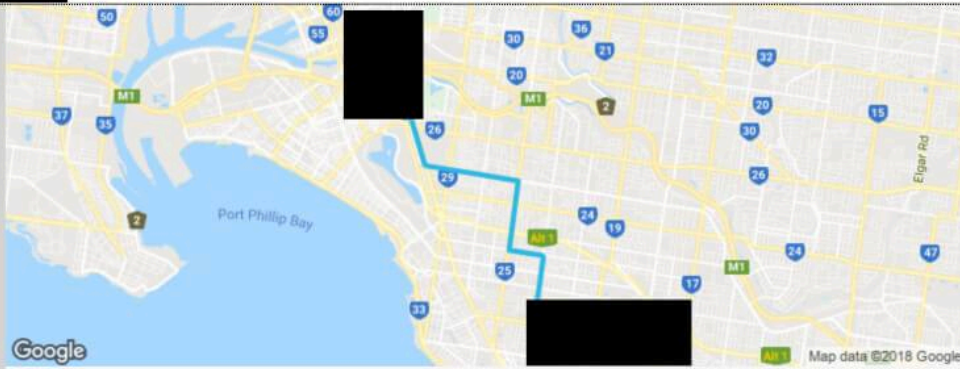
From Uber Receipts <uber.australia@uber.com> ☆

Subject Your Saturday evening trip with Uber

Reply Reply All Forward Archive Junk Delete More

14/7/18, 19:52

To [REDACTED]



A\$30.46

Thanks for choosing Uber, [REDACTED]

July 14, 2018 | UberX

07:28pm | [REDACTED] Australia

07:52pm | [REDACTED] Australia

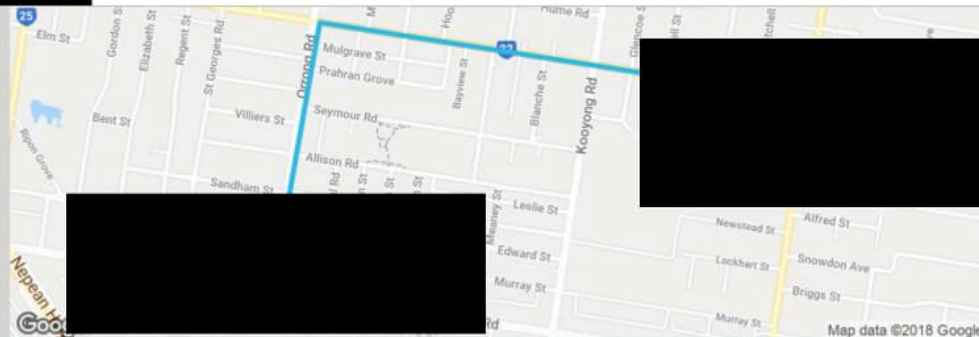
From Uber Receipts <uber.australia@uber.com> ☆

Subject Your Saturday evening trip with Uber

Reply Reply All Forward Archive Junk Delete More

4/8/18, 19

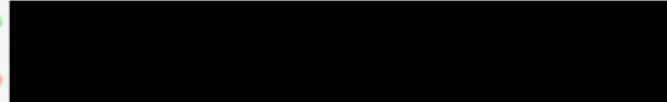
To [REDACTED]



A\$12.96

Thanks for choosing Uber, [REDACTED]

August 4, 2018 | UberX



UBER

They order things from Amazon:

From Amazon.com.au <digital-no-reply@amazon.com.au> ☆

Subject: Your Amazon.com.au Order What Alice Forgot.

To [REDACTED]

Reply

Reply All

Forward

Archive



[Your Account](#) | [Amazon.com.au](#)

Order Confirmation

Order [REDACTED]

Hello [REDACTED]

Thank you for shopping with us. All Kindle content, including books and Kindle active content, that you've purchased from the Kindle Store is stored in your [content library](#) on Amazon.com.au.

View and manage your books from your content library.

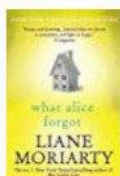
[Manage Your Kindle](#)



Order Details

Order [REDACTED]

Placed on Sunday, August 12, 2018



[What Alice Forgot](#)

Kindle Edition

Sold by Macmillan (AU)

[Review this item](#)

\$9.99

Item(s) Subtotal: **\$9.99**

Order Grand Total: \$9.99

Lawyers tend to use lots of mobile data (bonus for the emails revealing the active phone numbers of former staff):

From Telstra Mobile Data Usage <mobiledatausage@telstra.com> ☆

Subject: Mobile Data Group Usage Notification from Telstra

To [REDACTED]

Reply

Reply All

Forward

Archive

Junk

Delete

More

09:08



Mobile Data Usage Notification

Hello,

We wanted to let you know you've used 85% of your 2.60GB included shared data allowance for the mobile service 04 [REDACTED] with 12 days left.

Keeping you informed

Phone number

We'll continue to notify you by email once you reach 100% of your monthly shared data allowance even if you've added a Data Pack to your current plan.

From: Optus Mobile Data <wireless@optusnet.com.au> ☆

Subject: Your Optus Mobile Broadband usage alert

To: [REDACTED]

Reply

Reply All

Forward

Archive

Junk

Delete

More

14:47



Hi there,

Just letting you know you've used more than 24.26GB of your 48.50GB monthly mobile broadband data inclusion. Keep in mind itâ€™s for usage in Australia and the alert may be up to 48 hours old.

For your most up to date usage [login to My Account anytime](#).

Phone number

Service Number: 614 [REDACTED]

[Login to My Account](#)

What happens next?

Don't worry! We'll keep you updated as you use your service.

Then if you go over your 48.50GB data inclusion we'll automatically give you extra data in 1.00GB increments each charged at a flat rate of \$10.00 for use during the month. Your usage will reset on 27/07.

Cheers

The Optus Team

THIS IS A SYSTEM GENERATED EMAIL - PLEASE DO NOT REPLY TO THIS EMAIL

Text-to-email services leak text messages of personal nature:

From: 614 [REDACTED]@optusmobile.com.au ☆

Subject: [REDACTED]

To: [REDACTED]

Reply

Reply All

Hey

Tried to call a few times

[REDACTED]

Ironically, they receive invitations to cybersecurity events:

From: CLA <memberservices@cla.org.au> ☆
Subject: Choose one: upcoming CLA seminars in August and September
To: [REDACTED]
24/7/18, 15:26

Are there any defenses which can be raised to a claimed breach of s46?

How do the ACCC Interim Guidelines assist in the Interpretation of the new s46?

~~~~~

**How to protect yourself from evolving risks and cyber crime**

A commercial law forum  
5.30-7.00pm, Wednesday, 29 August, 2018  
The Royal Exchange of Sydney, 1 Gresham Place, Sydney  
**Free for financial CLA members, non-members \$49**

[Register Now](#)

Speakers:  
Nick Klein, Managing Director, Klein & Co Computer Forensics  
Katherine Sainty, Director, Sainty Law

Don't become the next cyber crime statistic. Arm yourself with key strategies to protect your business and clients against cyber threats.

The discussion will cover:  
**Privacy and data protection — what you need to know**  
**Developing cyber policies and procedures**

Finally, lawyers know how to party:

From: [REDACTED]  
Subject: CBD  
To: [REDACTED]  
4/7/18, 13:40

[REDACTED]

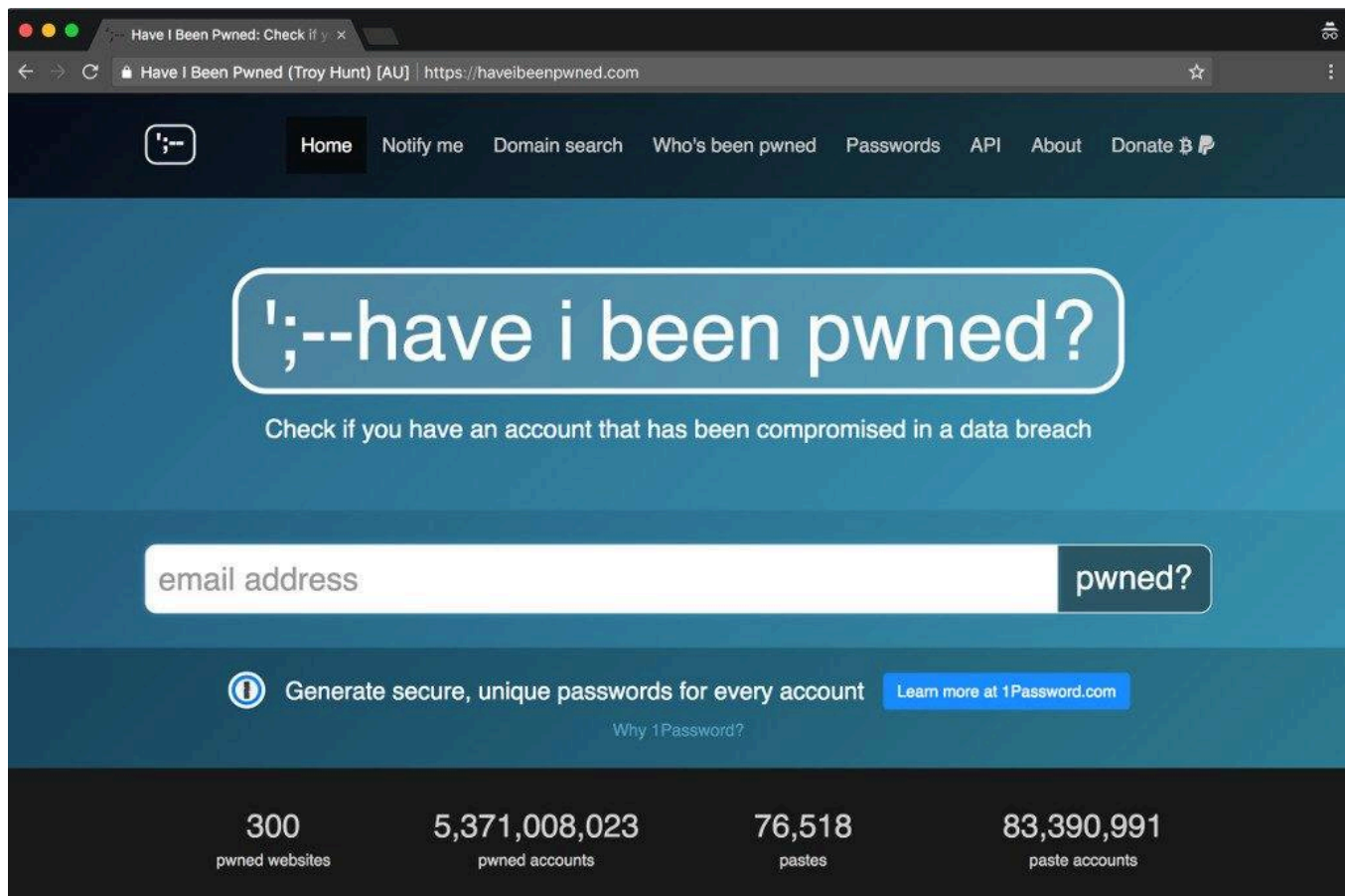
Good to catch up with you yesterday. Glad I left when I did though as you boys are league ahead of me on the drinking! Anyway I am free to catch up tomorrow. Just let me know when and what time works best. [REDACTED] around in the afternoon as well.  
Kind regards,  
[REDACTED]

## Revealing Valid Passwords from Data Breaches

In addition to setting up a catch-all email address, we took proactive steps to get to know our new domains better by registering to data breach notification websites. In doing so, we were able to reveal passwords belonging to legal professionals and staff at the former firms.

According to a recent study, [over 80% of people online are guilty of reusing their passwords on multiple cloud services](#). Passwords are often leaked to the internet when data breaches happen. There are [over 1.7 billion hacked credentials](#) from data breaches such as [LinkedIn](#), [Netflix](#) and [Adobe](#). If a legal practitioner is reusing the same password across several websites (such as their work or personal mailbox) as in the breach data, a hacker could log into their email service with the same password.

On [Haveibeenpwned](#) and [SpyCloud](#), email and domain name owners can check if they have an account that has been compromised in a data breach. It usually means that passwords from online services have ended up on the internet for everyone to see.

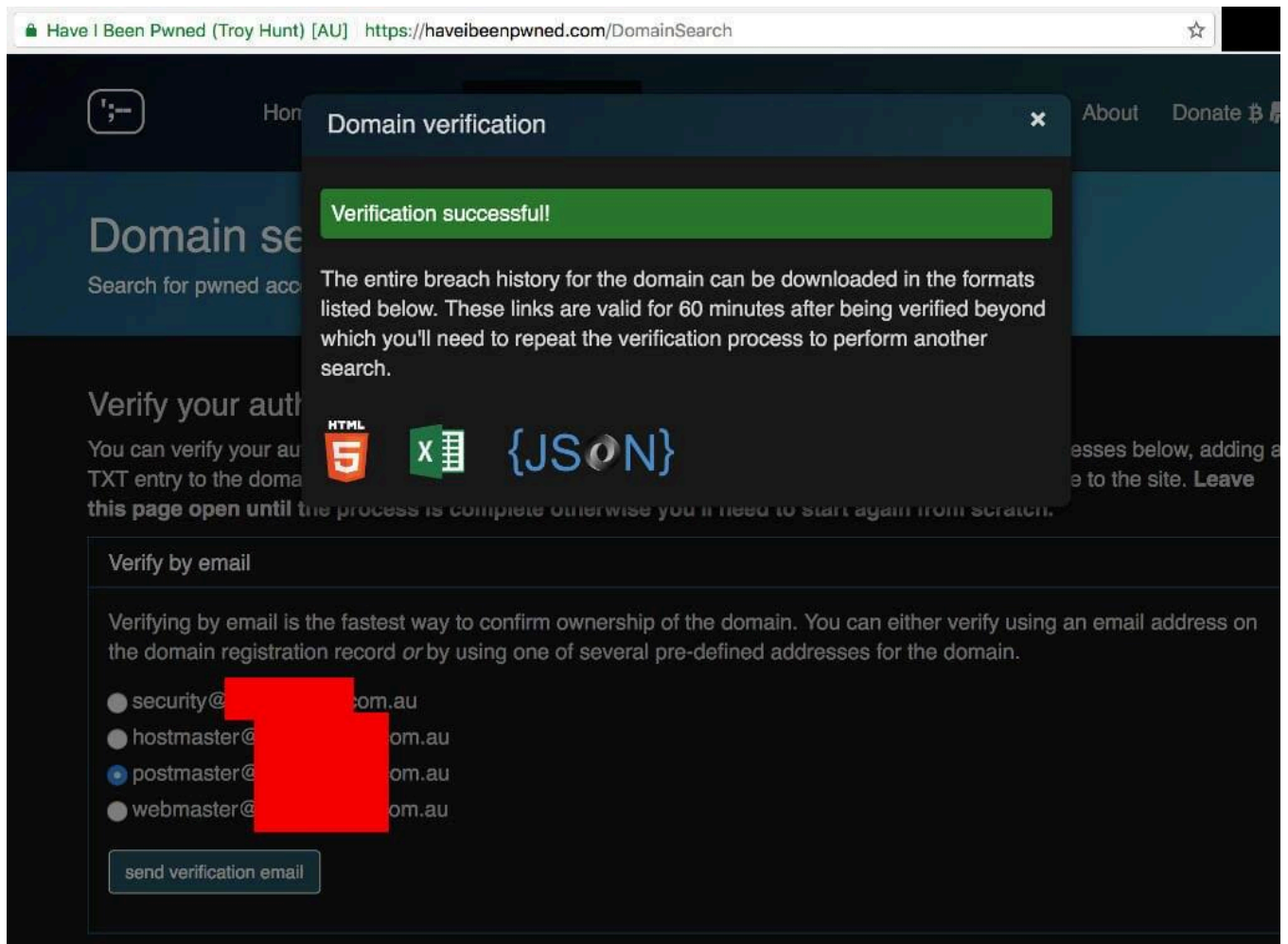


The screenshot shows the homepage of the 'Have I Been Pwned' website. The browser's address bar displays 'Have I Been Pwned (Troy Hunt) [AU] | https://haveibeenpwned.com'. The navigation bar includes links for Home, Notify me, Domain search, Who's been pwned, Passwords, API, About, and Donate. The main heading is 'have i been pwned?' with a subtitle 'Check if you have an account that has been compromised in a data breach'. Below this is a search form with a text input labeled 'email address' and a button labeled 'pwned?'. A promotional banner for 1Password is visible, stating 'Generate secure, unique passwords for every account' with a link to 'Learn more at 1Password.com'. The footer displays four statistics: 300 pwned websites, 5,371,008,023 pwned accounts, 76,518 pastes, and 83,390,991 paste accounts.

| Category       | Count         |
|----------------|---------------|
| pwned websites | 300           |
| pwned accounts | 5,371,008,023 |
| pastes         | 76,518        |
| paste accounts | 83,390,991    |

With the combination of the [Haveibeenpwned Domain Search](#) and the SpyCloud service, we were able to retrieve former legal practice email addresses and passwords leaked by past data breaches. Both of these services required us to verify the domain ownership before they provided access to the breach data information, but because we had full control over the domain names, we could easily pass this domain ownership verification process.

At Haveibeenpwned, we simply requested the confirmation email to *[[email protected]]* (<https://blog.gaborszathmari.me/cdn-cgi/l/email-protection>) to complete the verification process.



Once the verification was complete, we could retrieve the list of email addresses that were involved in any past data breaches.

|                                                                              |                                                                                                                       |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Have I Been Pwned (Troy Hunt) [AU]   https://haveibeenpwned.com/DomainSearch |                                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED] au                                                                | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED] m.au                                                              | Exploit.In, LinkedIn, Onliner Spambot                                                                                 |
| [REDACTED] n.au                                                              | Onliner Spambot                                                                                                       |
| [REDACTED] m.au                                                              | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | 2,844 Separate Data Breaches, Anti Public Combo List, Dungeons & Dragons Online, Exploit.In, MySpace, Onliner Spambot |
| [REDACTED] m.au                                                              | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED] m.au                                                              | Onliner Spambot                                                                                                       |
| [REDACTED] m.au                                                              | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED]                                                                   | Onliner Spambot                                                                                                       |
| [REDACTED] com.au                                                            | Onliner Spambot                                                                                                       |
| [REDACTED]                                                                   | Onliner Spambot                                                                                                       |

*This law firm had many employees whose emails were involved in a data breach*

The verification process at *SpyCloud* *\*was similar, all we had to do was click on a link in a domain ownership verification email. As opposed to \*Haveibeenpwned, however, this service exposes the actual passwords of the former employees, not just whether they were involved in a data breach.*

Secure <https://portal.spycloud.com/watchlist/identifiers/view/> show-tab-corporat...

SpyCloud

Data for [redacted] .com.au

Data records matching this email.

Filter Show Passwords ☒

Email Records (6) Potential Keylogger Records (0)

Show 10 entries

Column visibility Search:

| Breach Title                              | SpyCloud Publish Date | Breach Date | Email      | Password   | Sighting |
|-------------------------------------------|-----------------------|-------------|------------|------------|----------|
| 1M Mixed Combolist                        | 2018-05-08            | 2018-04-19  | [redacted] | [redacted] | 1        |
| Combolist of 1.4 Billion Credentials      | 2017-12-22            | 2017-12-22  | [redacted] | [redacted] | 4        |
| Anti-Public Combo list                    | 2017-10-18            | 2017-10-19  | [redacted] | [redacted] | 3        |
| Exploit.in                                | 2017-10-09            | 2017-10-07  | [redacted] | [redacted] | 2        |
| Dungeons & Dragons Online (February 2013) | 2017-03-31            | 2015-07-03  | [redacted] | [redacted] | 1        |
| MySpace                                   | 2016-10-21            | 2016-05-27  | [redacted] | [redacted] | 1        |

Showing 1 to 6 of 6 entries

We could reveal the actual passwords of legal professionals on SpyCloud

**Without publishing the actual passwords as part of this research, we can reveal that legal professionals (in our non-representative sample of thirty-something individuals) are:**

- *\*guilty of using weak passwords on online services; and \**
- **tend to reuse them across multiple websites.**

Because legal professionals tend to reuse their favourite passwords, it is likely that they chose the same favourite password on:

- their current business mailbox;
- their personal mailbox; or
- online services (e.g. Dropbox, OneDrive, Facebook).

## Abusing Password Resets of Social Media Accounts

By having the list of valid email addresses taken from Haveibeenpwned, we can demonstrate how we could have taken control over the current personal and work-related user accounts of former staff.

For example, practitioners tend to feature their former work email address on LinkedIn. Perhaps it is a little-known fact that everyone can request passwords reset emails to any of the email addresses added to the account.

Secure <https://www.linkedin.com/psettings/email>

Back to LinkedIn.com

in

Gabor, you're the boss of your account.

 **Gabor Szathmari**  
Director and cyber security expert at Iron Bastion  
Member since [redacted] [redacted] connections

Account Privacy Ads Communications

Login and security

Site preferences

Subscriptions and payments

Partners and services

Account management

### Login and security

#### Email addresses

Add or remove email addresses on your account Close

5 email addresses

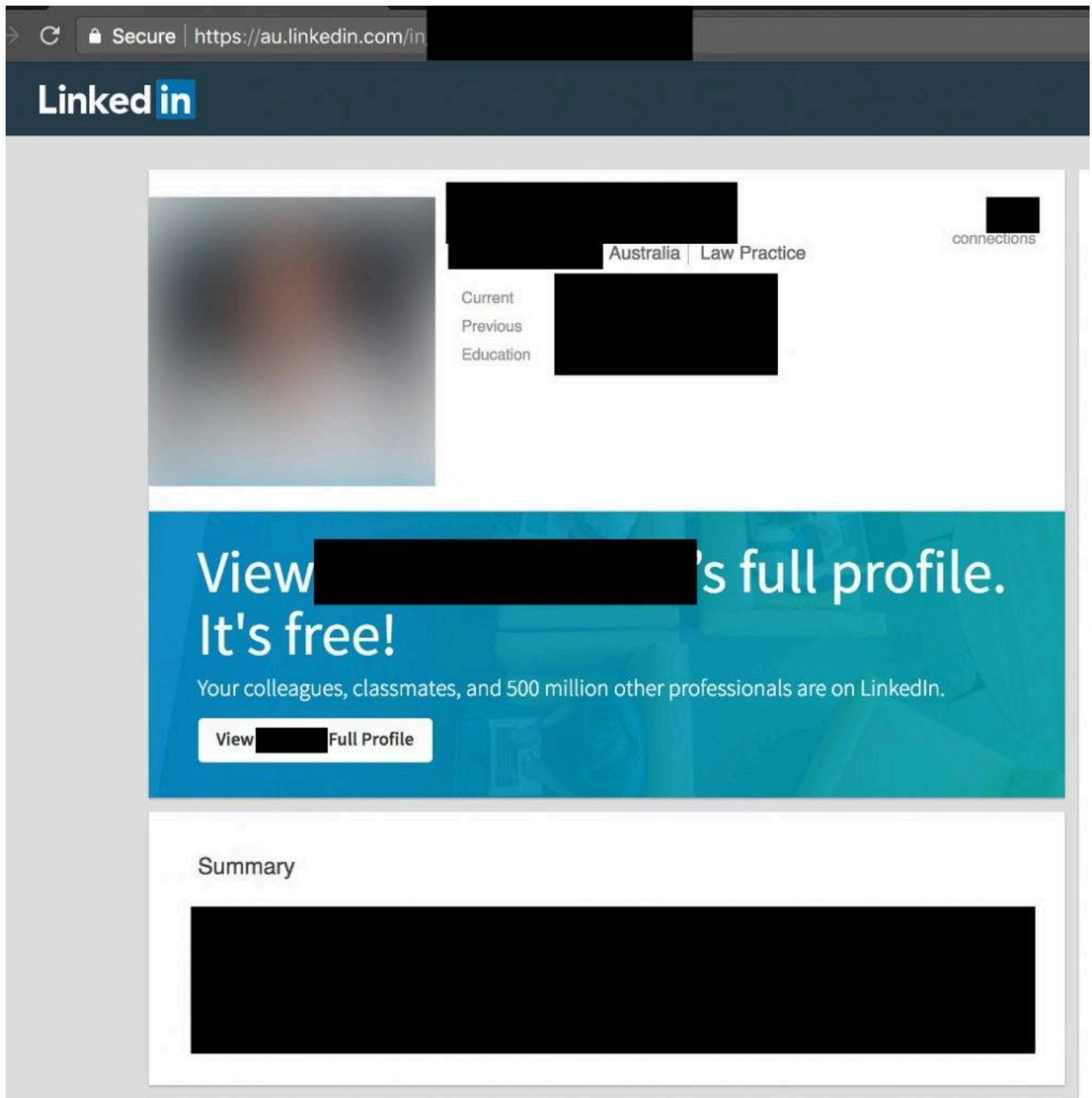
Email addresses you've added:

|            | Primary                                             |
|------------|-----------------------------------------------------|
| [redacted] | <a href="#">Make primary</a> <a href="#">Remove</a> |
| [redacted] | <a href="#">Make primary</a> <a href="#">Remove</a> |
| [redacted] | <a href="#">Make primary</a> <a href="#">Remove</a> |

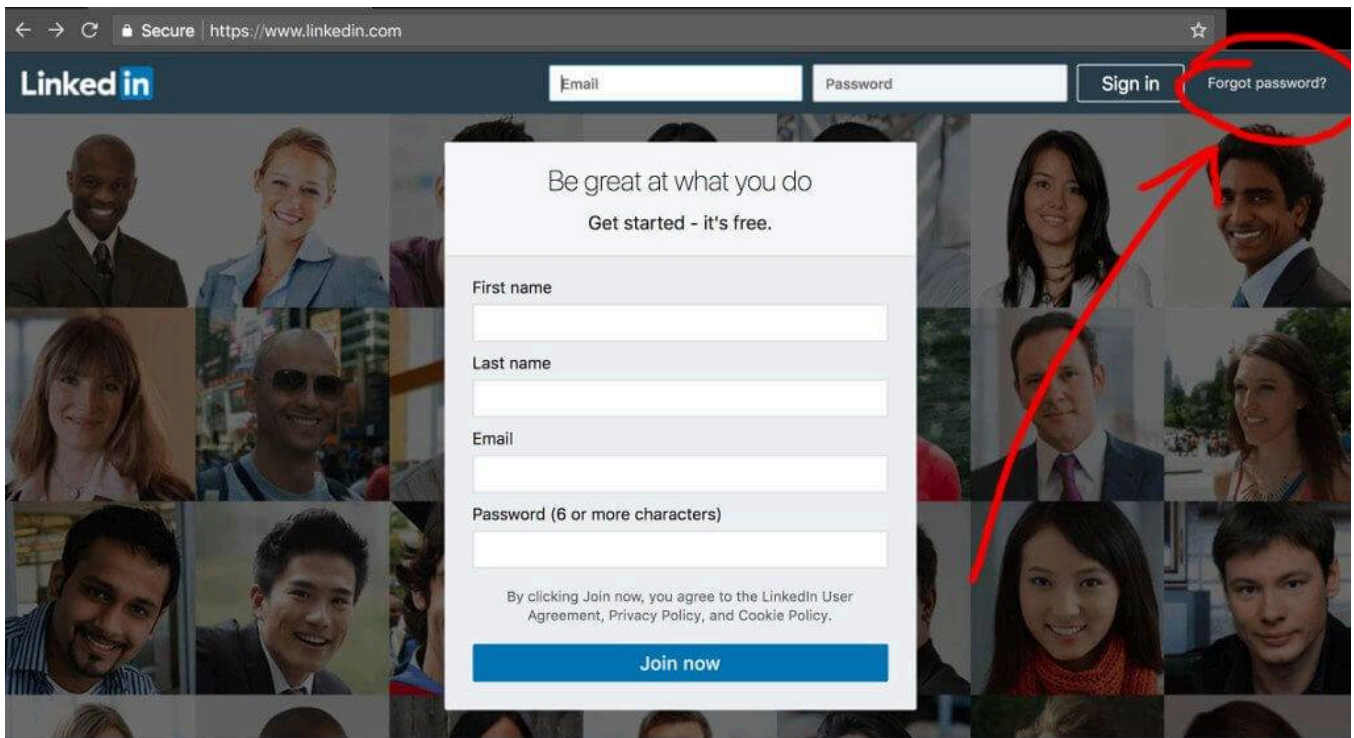
## Resetting passwords on personal LinkedIn accounts

Because email addresses associated with the abandoned domain names rarely get removed from the practitioners' LinkedIn account, we can request password reset emails to the domain under our control. All we need to do is go through the [LinkedIn Password Reset](#) process and click on the link in the email to hijack the practitioner's LinkedIn account.

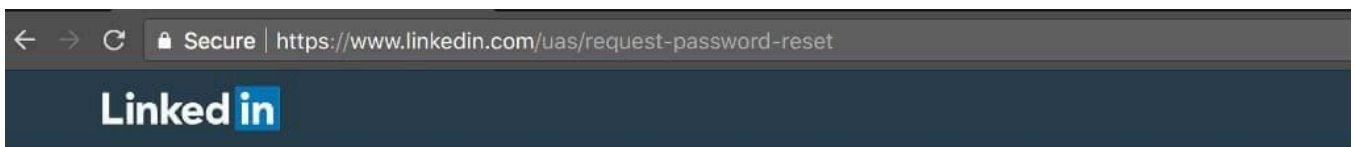
The following legal practitioner with a very active LinkedIn account was a partner at the former legal firm:



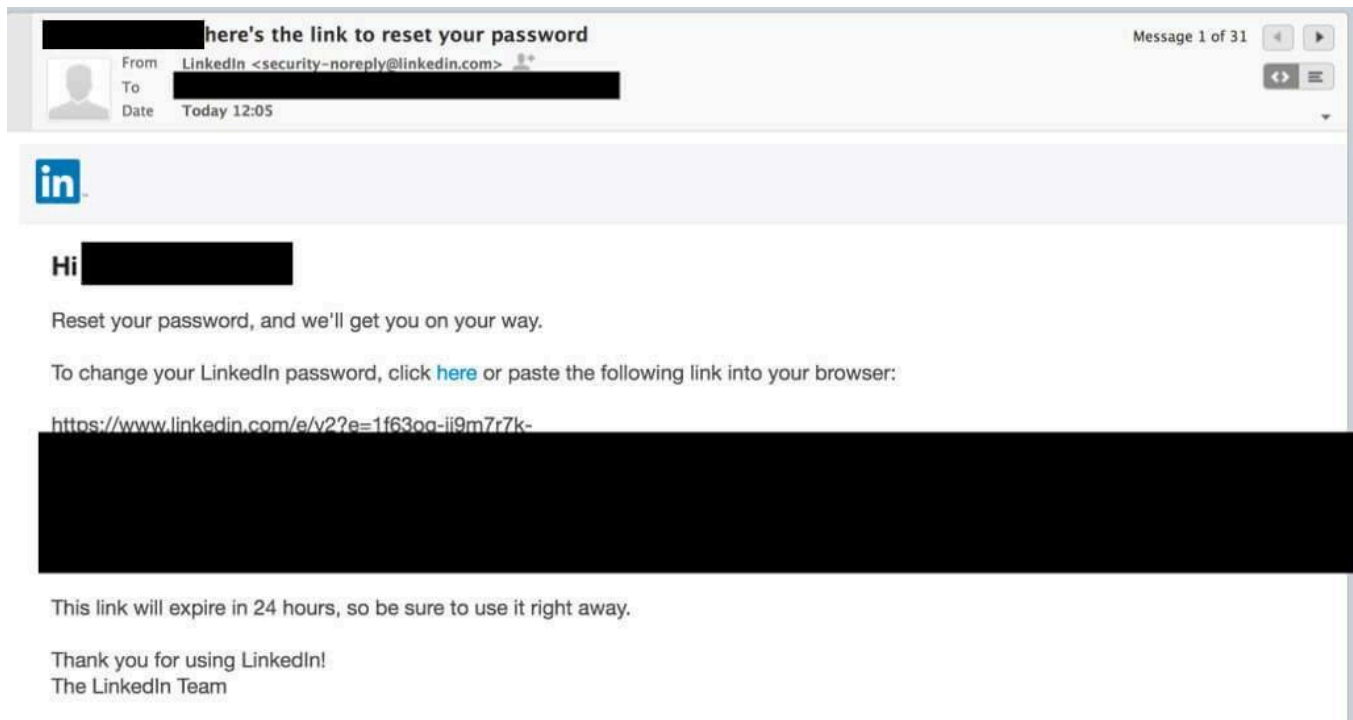
The first step is to visit the 'Forgot password?' page linked from the login page:



Next, we enter the practitioner's abandoned business email address:



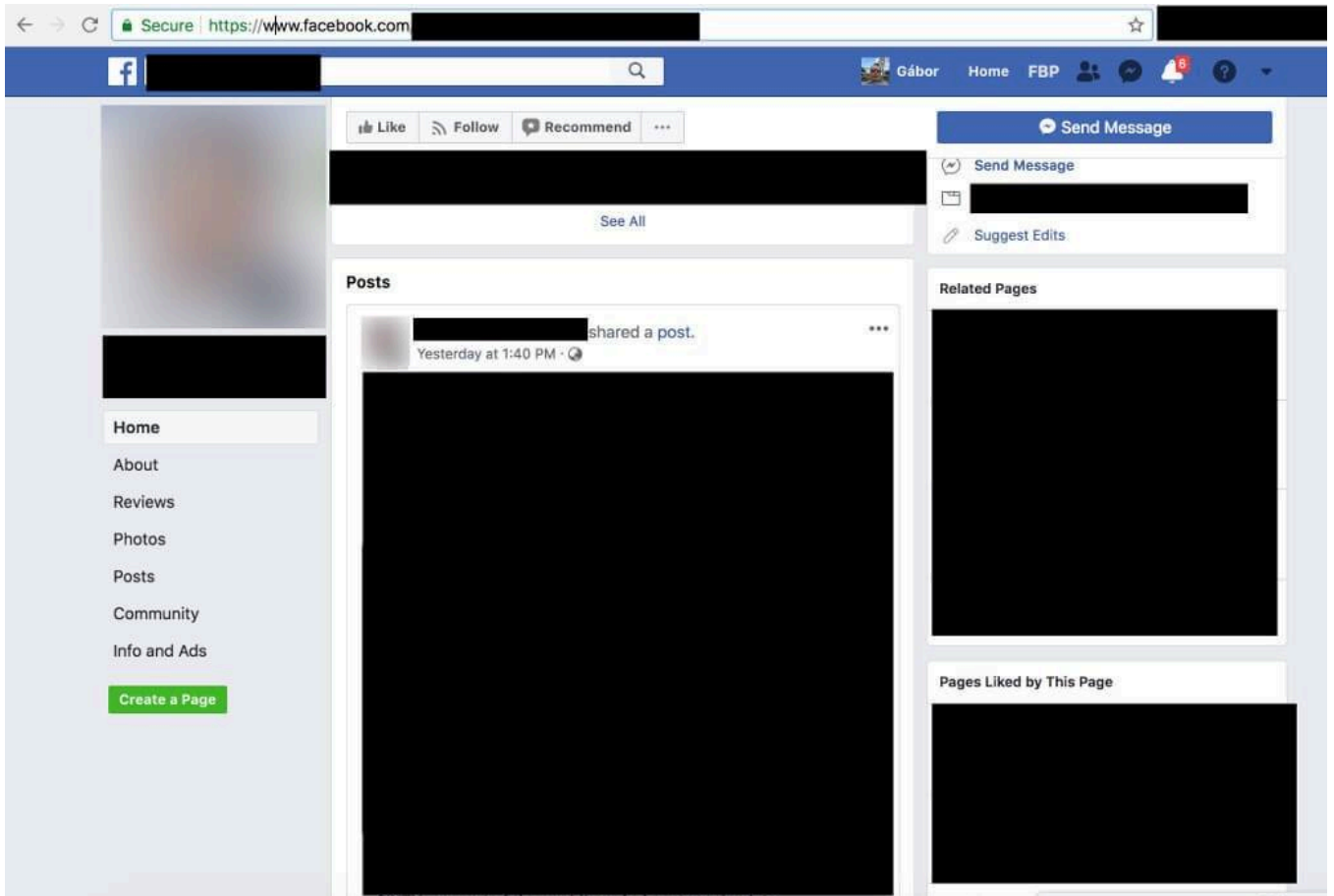
Then we receive the password reset email:



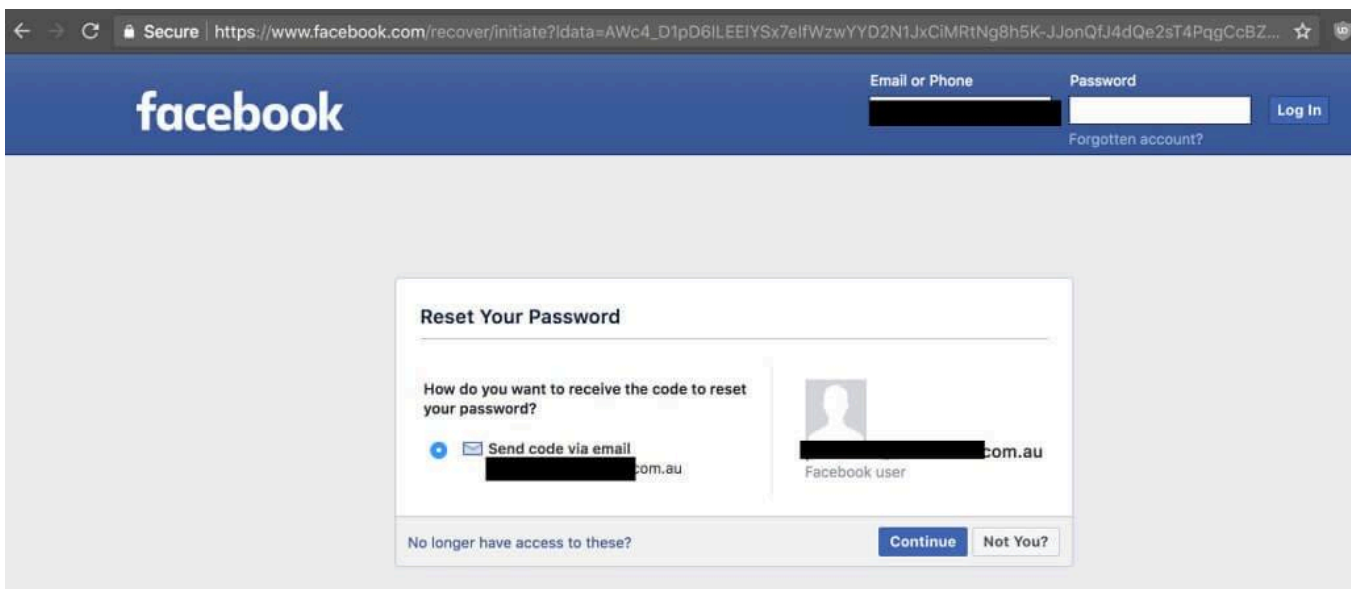
## Personal Facebook accounts

The same concept applies to Facebook as well. Certain practitioners also added their former work email address to their Facebook account and forgot to remove them. This practice allows us again to reset the password on Facebook, too.

The following solicitor owns a quite active Facebook page:



Let's find out if we can reset this solicitor's password with the 'Forgotten account?' feature:



The password reset email arrives as expected:

From Facebook <security@facebookmail.com> ☆  
Subject **784561 is your Facebook account recovery code**  
Reply to noreply <noreply@facebookmail.com> ☆  
To [REDACTED]

18:35



Hi [REDACTED]

We received a request to reset your Facebook password.

[Click here to change your password.](#)

Alternatively, you can enter the following password reset code:

784561

**Didn't request this change?**

If you didn't request a new password, [let us know](#).

[Change Password](#)

This message was sent to [REDACTED] your request.  
Facebook, Inc., Attention: Community Support, 1 Facebook Way, Menlo Park, CA 94025

We could just use the embedded link then or provide the six-digit reset code on the website to complete the account takeover:

Secure https://www.facebook.com/recover/code/?em[0]: [REDACTED]

**facebook**

**Enter Security Code**

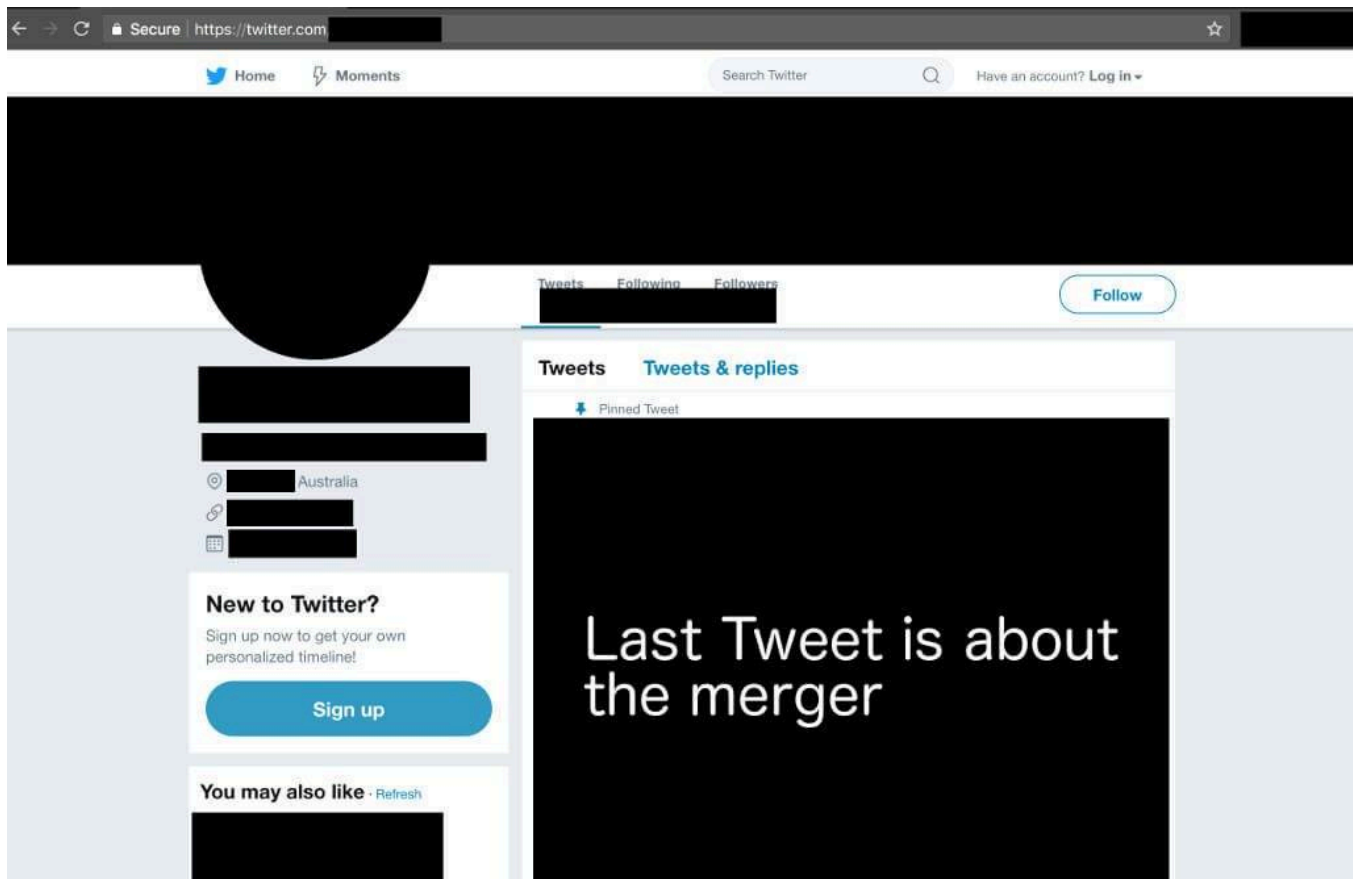
Please check your email for a message with your code. Your code is 6 digits long.

We sent your code to: [REDACTED]om.au

[Didn't get a code?](#) [Continue](#) [Cancel](#)

## Accessing Twitter accounts

Twitter is no exception either. We found Twitter accounts registered under someone's former email address under the abandoned domain, making Twitter accounts susceptible to password resets.



We use the forgotten password feature again:



The link in the email would allow us to reset the password of the Twitter account and let us in:

## Check your email

We've sent an email to  Click the link in the email to reset your password.

If you don't see the email, check other places it might be, like your junk, spam, social, or other folders.

[I didn't receive the email](#)

From Twitter <password@twitter.com> 

Subject **Password reset request**

To 

 Reply

 Reply All ▾

 Forward

 Archive

 Junk

 Delete

 More ▾

19:03



### Reset your password?

If you requested a password reset for  click the button below. If you didn't make this request, ignore this email.

Reset password

#### Getting a lot of password reset emails?

You can change your [account settings](#) to require personal information to reset your password.

#### How do I know an email is from Twitter?

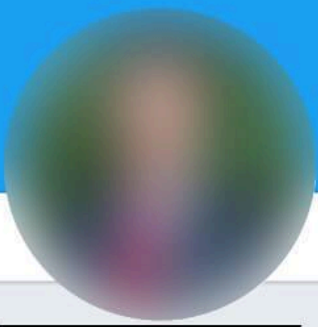
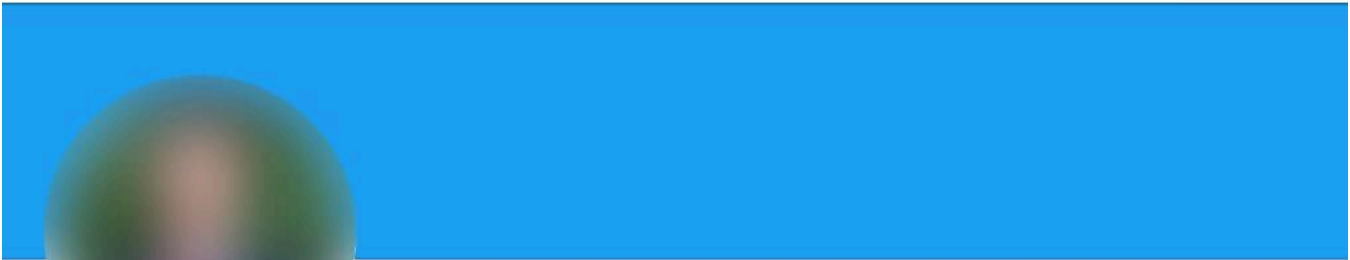
Links in this email will start with "https://" and contain "twitter.com." Your browser will also display a padlock icon to let you know a site is secure.

[Help](#) | [Not my account](#) | [Email security tips](#)

This email was meant for 

Twitter, Inc. 1355 Market Street, Suite 900 San Francisco, CA 94103

Personal Twitter accounts are not safe, either, as certain practitioners used their work email address to register on Twitter and never changed it:



Tweets Following Followers Likes

Australia

Photos and videos

Tweets Tweets & replies Media

Retweeted

From Twitter <password@twitter.com> ☆

Subject Password reset request

To [REDACTED]

Reply

Reply All

Forward

Archive

Junk

Delete

More

21:15



## Reset your password?

If you requested a password reset for [REDACTED] click the button below. If you didn't make this request, ignore this email.

[Reset password](#)

### Getting a lot of password reset emails?

You can change your [account settings](#) to require personal information to reset your password.

### How do I know an email is from Twitter?

Links in this email will start with "https://" and contain "twitter.com." Your browser will also display a padlock icon to let you know a site is secure.

[Help](#) | [Not my account](#) | [Email security tips](#)

This email was meant for [REDACTED]  
Twitter, Inc. 1355 Market Street, Suite 900 San Francisco, CA 94103

## Abusing Profession-specific Services

Legal practitioners rely on free services like Dropbox for storing and sharing work-related files. This Dropbox account seems to be full per the notification email which landed in our mailbox:

From Dropbox <no-reply@dropbox.com> ☆

Subject Your Dropbox has stopped syncing

To [REDACTED]

Reply

Reply All

Forward

Archive

Junk

Delete

More

18/7/18, 02:27



Hi [REDACTED]

Your Dropbox is full and is no longer syncing files. New files added to your Dropbox folder won't be accessible on your other devices and won't be backed up online.

Upgrade your Dropbox today and get up to 1 TB (1,000 GB) of space and powerful sharing features.

[Upgrade your Dropbox](#)



Let's see if we could reset the password on it! (spoilers: yes, it would)

Dropbox, Inc [US] | [https://www.dropbox.com/forgot?email\\_from\\_login=](https://www.dropbox.com/forgot?email_from_login=)

[Try Dropbox Business](#)

If a Dropbox account exists for [REDACTED]om.au, an e-mail will be sent with further instructions.

[Log in](#) • [Create an account](#)

## Forgot your password?

Enter your email address to reset your password. You may need to check your spam folder or unblock no-reply@dropbox.com.

[REDACTED]om.au

[Submit](#)

[I can't recover my account using this page](#)

From Dropbox <no-reply@dropbox.com> ☆

↩ Repl

Subject **Reset your Dropbox password**

To [REDACTED]



Hi [REDACTED]

Someone recently requested a password change for your Dropbox account. If this was you, you can set a new password here:

Reset password

If you don't want to change your password or didn't request this, just ignore and delete this message.

To keep your account secure, please don't forward this email to anyone. See our Help Center for [more security tips](#).

Happy Dropboxing!

## Professional-specific web portals

The hijacked email addresses also allow us to reset the password of the **Commonwealth Courts Portal**. The Commonwealth Courts Portal provides web-based registry services for legal professionals to file documents for litigation process for the Federal Court. A user account here could give us access to sensitive documents and details of former clients.

Secure https://www.comcourts.gov.au/plp/saml/authn\_request?SigAlg=http%3A%2F%2Fwww.w3.org%2F2000%2F09%2Fxmldsig%23rsa-sha1&S...

Federal Circuit Court of Australia | Family Court of Australia | Federal Court of Australia

 Commonwealth Courts Portal [Federal Law Search](#)

## Welcome to the Commonwealth Courts Portal

The Commonwealth Courts Portal is an initiative of the Family Court of Australia, Federal Court of Australia and Federal Circuit Court of Australia. It provides web-based services for clients to access information about cases before the courts.

If you have already registered, please enter your user name and password to login

If you have not registered, you will need to [Register now](#)

[Federal Law Search](#) provides selected information on cases initiated in the Federal Court of Australia and in the federal law jurisdiction of the Federal Circuit Court of Australia.

Family Law eService [Obtain Proof of Divorce](#) or [Pay Hearing Fees](#) without registering and logging on to the Commonwealth Courts Portal.

### News and updates

#### New users

New users can [Register now](#)

#### Registered users

User name:

Password:

[Forgotten user name](#)  
[Forgotten password](#)

 OFFLINE

Although the portal requires a username and password combination to log in, we can retrieve the username by entering our email address:

Secure https://www.comcourts.gov.au/pacm/access/send\_username

Family Court of Australia | Federal Court of Australia | Federal Circuit Court of Australia

 Commonwealth Courts Portal [Home](#) [Federal Law Search](#)

### Forgotten user name

If you have previously registered for the Commonwealth Courts Portal and have forgotten your user name, enter the email address you provided when registering and your user name will be sent by email.

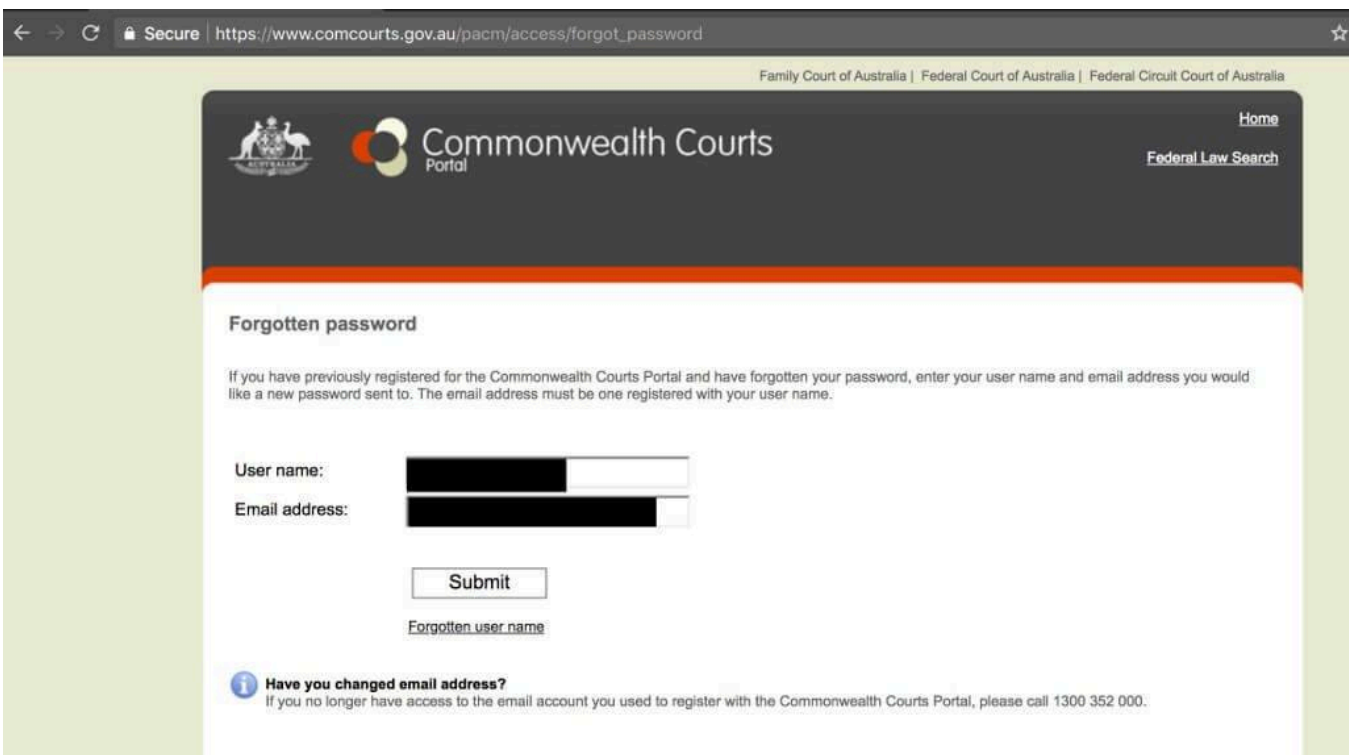
**Please enter your email address.**

Email:

 **Have you changed email address?**  
If you no longer have access to the email account you used to register with the Commonwealth Courts Portal, please call 1300 352 000.



We can use the forgotten password feature by keying in the username from the email and the very same email address.



The portal assigns us a temporary password, which would let us log in then:

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/comcourts-5.jpg)

We can also reset passwords on the **NSW Online Registry** portal, too. The Online Registry portal provides similar services to the Commonwealth Courts Portal, but for state courts such as the NSW Supreme, District and Local Courts.

We use the forgotten password feature again to get access:

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/nsw-online-registry-1.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/nsw-online-registry-2.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/nsw-online-registry-3.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/nsw-online-registry-4a.jpg)

Once we clicked on the password reset link from the email, we did not attempt to proceed past the security questions. However, as Google pointed out earlier, [security questions are insecure](#). The adventurous may want to search for these details in public records. For example, we were able to track down this particular lawyer's older brother on Facebook, whose birth date is probably available on the platform.

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/nsw-online-registry-5.jpg)

The **LEAP Practice Management Platform** is not safe, either. LEAP practice management software and is the most commonly used software for managing a legal practice. The platform contains online client files, legal documents and has integrated trust accounting and time billing.

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/leap-1.jpg)

We click on the 'Forgotten password?' link again and enter one of the legal practitioner's former work email address:

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/leap-4.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/leap-2.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/leap-3.jpg)

A few seconds later, we managed to receive the following email:

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/leap-5.jpg)

Although [LEAP is boasting how secure their platform is](#), the password reset email features the cleartext password, meaning that the company is not storing their customers' passwords in a [secure hashed format](#).

Finally, **Law Society accounts** are not safe from password reset attacks, either:

[[image: image]](https://blog.gaborszathmari.me/hacking-law-firms-abandoned-domain-name-attack/law-society-1/)

[[image: image]](https://blog.gaborszathmari.me/hacking-law-firms-abandoned-domain-name-attack/law-society-2/)

## Law firms also use PayPal

Certain firms had registered on PayPal with their work email as a method of receiving payments from clients:

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/paypal-1.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/paypal-2.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/paypal-3.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/paypal-4.jpg)

## Google Accounts

This particular firm had an AdWords account at Google. If we were wondering what keywords this firm was using on AdWords? We could have reset the password to find that out.

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-aw-1.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-aw-2.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-aw-3.jpg)

## Accessing Former Office 365 and G Suite Accounts

---

Based on our experience, the two most popular email platforms amongst law firms are Office 365 followed by Google G Suite. These cloud-based email services are often abandoned leaving online data intact, rather than the accounts closed. To make things worse, legal professionals tend to retain their emails forever, making those mailboxes fairly valuable to potential fraudsters operating [Business Email Compromise \(BEC\)](#) fraud.

Based on the historical DNS records, we found that one of the law firms relied on Office 365 and G Suite for hosting email services. This made us think: could we hijack the account and access the inboxes of the former practice?

First, we tried and failed to reset the password on Office 365 as two-factor authentication was enabled, which stopped us from completing the password reset.

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/office365-1.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/office365-2.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/office365-3.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/office365-4.jpg)

[[image: image]](https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/office365-5.jpg)

We had more luck with G Suite. First, we tried and failed to reset the password with the former G Suite administrator's email account:

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-1.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-2.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-3.jpg>)

Then we tried to reset the G Suite administrator's account by using the internal email address that Google assigns to every subscriber.

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-4.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-5.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-6.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-7.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-8.jpg>)

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-9.jpg>)

We stopped at this last step and decided to not complete the password reset process on G Suite:

[[image: image]](<https://cdn9.gaborszathmari.me/wp-content/uploads/2018/08/google-10.jpg>)

As for all other services, we did not complete the final step of the password resets for privacy reasons meaning we did not log into or take over the user accounts, or access any information stored in online services, although we could have.

## Conclusion

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Businesses, especially legal practices leave themselves exposed to cyber attacks by allowing their former domain names expire. Bad actors can acquire these abandoned domain names and reinstate the former business's email service.

This research demonstrates that abandoned domain names allow new domain owners to access financial, personal, confidential and privileged information of the former owner. In addition, attacks can gain access to email addresses and passwords from past data breaches, and take over online services. If we were a bad actor, we could have used the

domain to commit fraud by numerous methods as well as reinstating the former website of the law firm and posing as former staff.

To prevent this from happening to your business, we recommend you:

- keep renewing the former firm's domain name indefinitely;
- [close user accounts](#) that were registered with the business email address (e.g. Dropbox, Commonwealth Courts Portal, PayPal);
- change or remove the business email address from online user accounts (e.g. LinkedIn, Facebook);
- [unsubscribe from email notifications](#) that usually features sensitive data (Text-to-email services, mobile phone billing notifications);
- [advise your clients to update their address book](#);
- [enable two-factor authentication](#) (2FA or MFA) where the feature is supported for online services; and
- [use unique and complex passwords](#).

We recommend that LEAP review the password storage practices of their practice management software and apply the latest [password hashing security practices](#). Online court portals and other professional-specific websites should implement two-factor authentication for logins and strict controls for password resets.

We also suggest the Australian law societies consider taking over the domain names when a legal practice is wound-up. As far as we know, law societies in Australia have the power to appoint an administrator to take over a legal practice when it is closed to take care of client files and distribute any funds left in the trust account. Law societies could take over the domain name and hold onto that for an extended period rather than letting them expire. The law societies should set up a website with a simple notice (like the FBI does on [seized domain names](#)) advising the visitors that the law firm is closed and reply to emails with an automated message.

During the three month period of this research, we:

- re-registered six abandoned domain names, some of which formerly belonged to Australian legal practices;
- received approximately 25,000 emails in total;
- received emails and documents of a sensitive nature;
- recovered the actual passwords (previously exposed in public data breaches and were later published on Spycloud) of approximately thirty legal professionals;
- successfully attempted password recovery of many popular online services and profession-specific portals;
- won \$250,000 from Mark Zuckerberg himself (we are yet to claim the prize).

## About the Authors

*Gabor Szathmari is a cybersecurity expert with over ten years experience, having worked in both private and public sectors. He has helped numerous big-name clients with data breach investigations and security incident management. In his professional life, Gabor*

*helps businesses, including many small and mid-size legal practices, with their cybersecurity challenges at [Iron Bastion – Australia’s anti-phishing experts](#). Gabor is a security consultant at Arrow Networks, a premier [managed IT services provider in Sydney](#). [Jeremiah Cruz](#) is a Networking Associate and UTS Graduate. He helps kids learn to code and communicates complex ideas through stories and practical lessons building what he most loves: Games.*

[image: image]

## **Gabor**

Gabor Szathmari is a cybersecurity expert and digital privacy enthusiast. In his professional life, Gabor helps businesses, including many small and mid-size legal practices, with their cybersecurity challenges at Iron Bastion.

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